Frobenius ; Fuchs

Diff : Integral Eqn's as examples of linear equations on spaces of functions:

$$\underset{D \;\; -I}{\left[\frac{d}{dx} \;\; -1\right]} u(x) = 0$$

$$A \qquad u \quad = 0$$

$\Rightarrow u \in$ [Null space of A] $:= N(A)$

$\Rightarrow u, v \in N(A)$, $\quad A(\alpha u + \beta v) = \alpha \underset{\| \; 0}{Au} + \beta \underset{\| \; 0}{Av}$

$= 0 + 0$

$\alpha u + \beta v \in N(A)$

The null space of A is a vector space.

Constructed polynomial approximations:

$$(**) \quad \underset{n \times n+1}{\begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ -1 & 2 & 0 & \cdots & 0 \\ & \ddots & \ddots & \ddots & \vdots \\ & & & & 0 \\ & & & & -1 \; n \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix}} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

boundary condition →

$u(1) = 1$

$\downarrow$

$u(x) = e^x$

$$u(x) \approx a_0 \sum_{k=0}^{n} \frac{1}{k!} x^k \xrightarrow{n\to\infty} a_0 \sum_{k=0}^{\infty} \frac{1}{k!} x^k = a_0 e^x$$

"evaluate" at $x = 1$

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n$$

$$p(1) = a_0 + a_1 + \cdots + a_n$$

Modified Power Series

ODE solutions are not always as "smooth" as a power series, and we need to modify our approach to "see" these singularities.

Euler's Equation

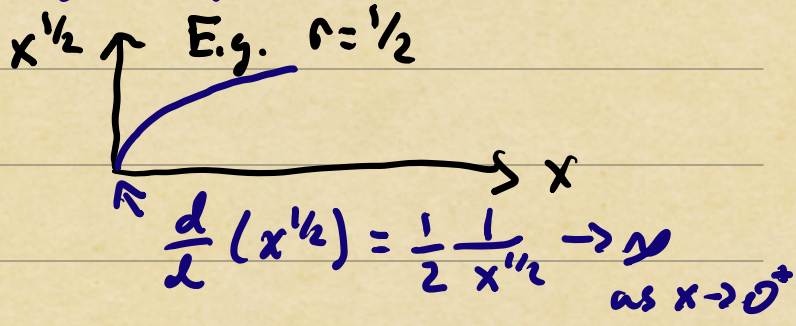
"singular" at $x=0$ ↓ numbers ↓ ↓

$$x^2 u''(x) + a x u'(x) + b u(x) = 0$$

$$A u = 0$$

"The origin ($x=0$) is a singular point of this ODE"

$$\tilde{u}(x) = x^r$$



"Indicial equation"

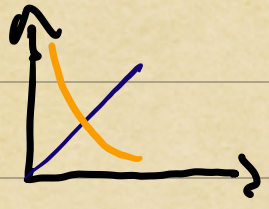
$$[r(r-1) + ar + b]\tilde{u}(x) = 0$$

$$[r^2 + (a-1)r + b]\tilde{u}(x) = 0$$

$$\Rightarrow \quad r_\pm = \frac{1}{2}\left(-(a-1) \pm \sqrt{(a-1)^2 - 4b}\right)$$

Solutions are <u>not</u> always smooth!

For $a-1 = 1/2$, $b = -1/2$ $\Rightarrow$ $r = 1/2, -1$



$$\Rightarrow u(x) = c_1 x^{-1} + c_2 x^{1/2}$$

In general, indicial equation determines singularity

$\Rightarrow$ If $\mathrm{Re}(r) \geq 0$, then the soln is b'dd at $x=0$

$\Rightarrow$ If $\mathrm{Re}(r) < 0$, then the soln's can blow up

$\Rightarrow$ If $\mathrm{Im}(r) \neq 0$, the soln's oscillate

$\Rightarrow$ Repeated roots, 2nd solution has a log term

$$u(x) = c_1 x^r + c_2 x^r \log x$$

Modified Power Series

<u>Idea:</u> Modify basis by "factoring" out the singular behavior, learned from an indicial equation

$$\{1, x, \dots, x^n\} \Rightarrow \{x^r, x^{r+1}, \dots, x^{r+n}\}$$

$x^r 1,\ x^r x,\ \dots, x^r x^n$

Equivalent to ansatz: $\tilde{u}(x) = x^r \sum_{n=0}^{\infty} a_n x^n$

(x^r: singular part; $\sum_{n=0}^{\infty} a_n x^n$: smooth power series)

$\Rightarrow$ Need to find power series coeffs & exponent

Example: $x^2 u''(x) + 4x u'(x) + (x^2+2)u(x) = 0$

($x = 0$: singular point)

$$\tilde{u}(x) = \sum_{n=0}^{\infty} a_n x^{n+r} \qquad (x^2+2)\tilde{u}(x) = \sum_{n=0}^{\infty} a_n x^{n+r+2} + \sum_{n=0}^{\infty} 2a_n x^{n+r}$$

$$\tilde{u}'(x) = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1} \qquad 4x u'(x) = \sum_{n=0}^{\infty} 4a_n (n+r) x^{n+r}$$

$$\tilde{u}''(x) = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-2} \qquad x^2 u''(x) = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r}$$

$\Rightarrow$ match coeffs/coords of x^{n+r} $(n \geq 0)$

$$\sum_{n=0}^{\infty} (\quad\quad) x^{n+r} = 0$$

$(\quad\quad) = 0$

Since x^{n+r} are lin. indep. functions

Convenient to organize matching in table:

coeffs of	x^r	x^{r+1}	x^{r+2}	...	x^{r+m}
x^2u''	$r(r-1)a_0$	$(r+1)ra_1$	$(r+2)(r+1)a_2$		$(n+r)(n+r-1)a_n$
$4xu'$	$4ra_0$	$4(r+1)a_1$	$4(r+2)a_2$		$4(n+r)a_n$
x^2u	0	0	a_0		a_{n-2}
$2u$	$2a_0$	$2a_1$	$2a_2$		$2a_n$

$$\Rightarrow \quad [r(r-1) + 4r + 2]a_0 = 0$$

$$\Rightarrow \quad [(r+1)r + 4(r+1) + 2]a_1 = 0$$

$$\Rightarrow \quad (n+r)(n+r-1)a_n + 4(n+r)a_n + a_{n-2} + a_n = 0$$